
Three experiments, and what each one actually settled

Look-ahead depth · optimizer family · the oracle instrument

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Exp1 Llama-2-7B / GPT-3.5 Exp2 Llama-3.2-1B 4-bit / gpt-4o-mini Exp3 Llama-3.2-1B bf16 / gpt-4o-mini, Claude Haiku 4.5 held out

Exp1 and Exp2 are PTO only. GRPO enters in Exp3.

Three experiments, and the question each one handed forward

I Exp1 — ICLR 2025

Established. PTO beats the untrained baseline, and deeper look-ahead looks better.

Left open. One model was patient, oracle AND training reward. Best-vs-best was chosen post hoc. The depth contrast is significant on one of four metrics.

II Exp2

Established. Harder patients, a JSON-schema oracle, three training instruments.

Left open. Only the Q1+Q2-trained arm moved. Training on WAI-SR or CSQ-8 did not improve WAI-SR or CSQ-8. Look-ahead was a wash at matched iterations.

III Exp3

Established. Look-ahead helps GRPO and hurts PTO; which optimizer wins depends on K — and both answers hold on a grader that was never the reward.

Left open. One arm stopped at iteration 6. Every contested endpoint is a single 96-conversation draw. Which instrument to TRAIN on is still untested here.

Every number that follows was read off a table or recomputed from raw data, then verified a second time against that source. Where one disagrees with a project doc, the slide says so.

What never changed across the three experiments

- **The task.** A therapist LLM runs a Motivational Interviewing session against a simulated patient. No human is in the loop at any point.
- **The personas.** 96 patients, factorial: gender × cooperation × problem × duration × prior attempts × age = 2·3·2·2·2·2. Identical construction in all three experiments, so every contrast is paired on persona, n = 96.
- **The reward.** A larger LLM grades the finished transcript on validated MI questionnaires. Q1 = session satisfaction (5 items); Q2 = working alliance (17 items).
- **The lever.** K = look-ahead depth. Before scoring a candidate reply, simulate K more turns and score where the conversation ENDS UP, not how the reply looks alone.

WHAT CHANGED, AND WHY

Exp1 → Exp2 one model played patient, oracle and reward. Split the patient off, harden it, and constrain the oracle to a JSON schema.

Exp1 → Exp2 7B → 1B therapist. Can a small model be taught this at all?

Exp2 → Exp3 add a SECOND grader that never played the patient and was never the reward.

Exp2 → Exp3 add GRPO, so PTO has something to be measured against.

Each experiment is a fresh re-implementation. No data flows between them.

ACT I

Exp1 — the published result

Llama-2-7B therapist, GPT-3.5 as patient, oracle and reward. PTO at $K \in \{0, 5\}$, 7 iterations each, 96 personas. Published at ICLR 2025.

What the paper reports

THE PUBLISHED CLAIM

Every PTO model beats the untrained baseline, and the K=5 models sit above the K=0 models on every metric in Table 1.

model	Q1	Q2	Final	Final SD	turns
Base (untrained Llama-2-7B)	3.521	3.385	3.453	0.740	43.7
K=0, best iteration (M4)	3.969	3.585	3.777	0.769	38.3
K=5, best iteration (M7)	4.190	3.775	3.982	0.414	34.4

Q1 = session satisfaction (5 items), Q2 = working alliance (17 items), both 1–5. Final = mean(Q1, Q2). n = 96 per row. Turn counts are from the prose, not a table — the paper publishes no length table.

— **All 15 rows reproduce exactly from the conversation data still in the repo.** 90 statistics, n = 96 each, matching to the third decimal (max deviation 0.0005). The measurement layer of Exp1 is sound — everything questioned on the next slide is in the inference layer.

What the paper's own statistics actually support

THE NARROWER TRUE CLAIM

PTO beats the baseline — that is solid and significant. But K=5 over K=0 is significant on ONE metric of four: working alliance, $p = 0.0315$.

metric	Base → K=0	Base → K=5	K=0 → K=5	p	significant?
Final Score	+0.324 **	+0.529 ***	+0.206	.0807	no
Q1 · satisfaction	+0.448 **	+0.669 ***	+0.221	.2095	no
Q2 · working alliance	+0.199 *	+0.390 ***	+0.191	.0315	YES
Conversation length	-5.34 *	-9.28 ***	-3.94	.0992	no

Tukey HSD, paper Table 4. * $p < .05$ ** $p < .01$ *** $p < .001$. The K=0 → K=5 column compares the two arms' best iterations, chosen after seeing the results.

— **The omnibus ANOVA was run on 3 groups, not 15.** Recomputing on Base + L0_M4 + L5_M7 reproduces all four published F-statistics exactly ($F = 15.637$ on Final). The honest all-15-model ANOVA gives $F = 4.299$ — still highly significant ($p = 1.6e-07$), so the conclusion survives; the presentation does not.

The same transcripts, re-scored four years later

IT REPLICATES — IN DIRECTION
Re-scoring all 1,440 Exp1 conversations with gpt-4o-mini keeps K=5 above K=0 at 7 of 7 iterations. The ordering is a property of the regime, not of GPT-3.5.

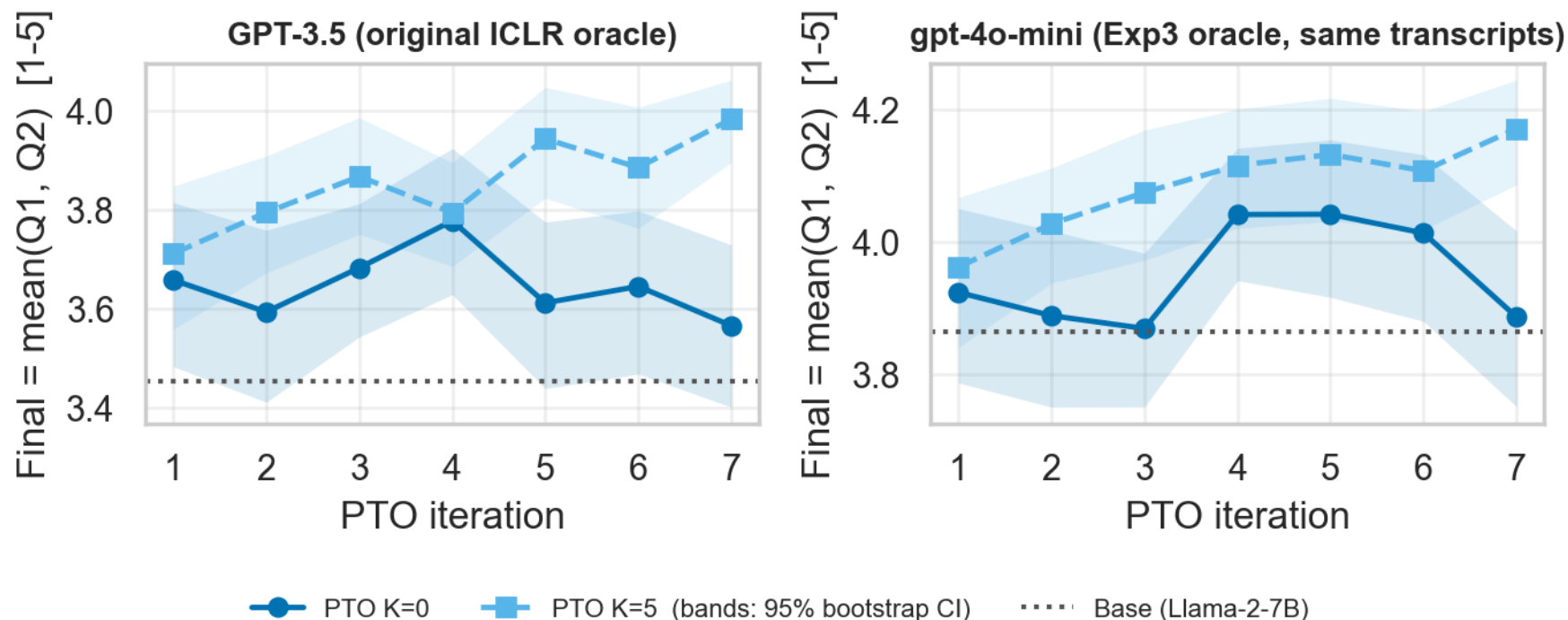
contrast (K=0 – K=5)	Δ modern	dz	p	Δ original	dz	p
Averaged over all 7 iterations	−0.132	−0.543	< .001	−0.206	−0.612	< .001
The paper's best-vs-best pair	−0.129	−0.250	.006	−0.206	−0.251	.331
Each arm at its own best	−0.129	−0.191	.405	−0.206	−0.251	.331

Metric = Final (mean of Q1, Q2). Negative = K=5 higher. Persona-paired, n = 96. 'modern' = the gpt-4o-mini re-scoring; 'original' = the GPT-3.5 scores Exp1 saved beside each conversation — not the thesis's held-out judge, which never saw Exp1. p = Wilcoxon signed-rank.

NOT THIS
Not that any single K=5 checkpoint beats any single K=0 checkpoint. Only the ARM-level average survives both graders with a CI clear of zero — and the paper's claim that every K=5 model outscores every K=0 model is false under BOTH graders (the first K=5 iteration breaks it).

Swapping the grader changes which arm has a result

Exp1 (ICLR 2025) conversations under two graders — K=0 solid, K=5 dashed



READ IT AS

Under GPT-3.5, 8 of 14 trained models separate from base after correction — including the K=0 arm's iteration 4, the paper's K=0 champion. Under gpt-4o-mini only 4 do, and ALL FOUR are K=5. The modern grader erases the K=0 arm's only significant gain.

The 'more stable' claim is a scoring ceiling

WHAT THE PAPER SAID

The best K=5 model has the lowest standard deviation on every metric — read as evidence that look-ahead produces more consistent therapists.

- **On Exp1's own numbers the ordering is real but not clean.** Lowest SD is indeed the K=5 endpoint (0.408). But the widest K=5 model (0.550) is wider than the tightest K=0 model (0.523) — the same first-iteration model that breaks the mean ordering. No variance test was run anywhere in the paper.
- **Re-tested properly on Exp3's 36 trained states, SD is predicted by the MEAN.** Spearman(mean, SD) = -0.863 , $p = 1.4e-11$ under the training oracle. Higher-scoring models are tighter because they are pressed against the top of a 1–5 scale.
- **The held-out judge, which has no ceiling, barely shows the effect.** Same correlation is only -0.376 . That gap is the evidence: Claude Haiku never awards ≥ 4.5 on Q1+Q2 in any of the 40 Exp3 model states, so it has nothing to compress against.
- **Concretely, at the best PTO state:** the cooperative third of personas scores 4.904 with SD 0.048 and 100% of conversations above 4.5. That is saturation, not stability. The resistant third of the same model still has SD 0.365.

What Exp1 settled, and what it handed forward

SETTLED

PTO trains a therapist LLM. Both arms beat the untrained baseline on every metric, with large and significant effects.

Look-ahead points the right way. K=5 sits above K=0 at 7 of 7 iterations, and it survives a grader swap four years later.

Trained models hold shorter sessions — 43.7 turns down to 34.4.

The data is trustworthy. Every published cell reproduces from the shipped files.

HANDED FORWARD

One model was patient, oracle and reward at once. There is no measurement independent of the thing being optimised.

K=5 over K=0 is significant on one metric of four, between two iterations picked after seeing the results.

The scale ceilings out. On cooperative personas the oracle has nowhere left to score.

It cannot be reproduced from its own text. The branching factor, trees per iteration and max length appear in the algorithms as symbols and are never given values; no DPO beta, learning rate, LoRA config, temperature or seed is stated anywhere.

A 7B therapist against cooperative patients is the easy regime. Does any of it hold when the model is smaller and the patients push back?

ACT II

Exp2 — the regime change

Llama-3.2-1B therapist in 4-bit. gpt-4o-mini as patient and oracle, on a JSON schema. Less cooperative patients. PTO across three training instruments $\times K \in \{0, 5\}$.

Four things changed at once

	Exp1	Exp2	the question it opens
Therapist	Llama-2-7B	Llama-3.2-1B, 4-bit NF4	can a 1B model learn this?
Patient	GPT-3.5, cooperative	gpt-4o-mini, resistant	remove the easy regime
Oracle	GPT-3.5, regex-parsed	gpt-4o-mini, JSON schema	stop losing scores to parsing
Instruments	Q1 + Q2	6 scored, 3 trained on	does the rubric choice matter?

- **Because four things changed together, Exp2 cannot attribute anything.** It is a regime change, not a controlled comparison — which is exactly why Exp3 holds everything fixed and moves one lever at a time.
- **Scale of the sweep:** 50 model states × 96 conversations × 6 questionnaires = 28,800 score files on disk, covering 4,800 scored conversations.

CORRECTION

Project docs describe Exp2 as “4 oracles × K” over “4,512 convs / 47 models”. Neither reproduces: only THREE training instruments exist on disk (Q1+Q2, WAI-SR, CSQ-8), and the census is 4,800 conversations across 50 model states.

PTO still works — but only when trained on Q1+Q2

THE RESULT

Every arm improves on the Q1+Q2 axis. Only the arm actually TRAINED on Q1+Q2 improves convincingly; the WAI-SR arms do not clear $p < .05$ at all.

trained on	K	best Q1+Q2	vs base	p	clears .05?
Q1+Q2	K=0	2.771	+0.393	.0006	yes
Q1+Q2	K=5	2.968	+0.590	<.0001	yes
WAI-SR	K=0	2.614	+0.237	.075	no
WAI-SR	K=5	2.590	+0.212	.096	no
CSQ-8	K=0	2.629	+0.251	.022	yes
CSQ-8	K=5	2.599	+0.221	.014	yes

Base = 2.378. Each arm's best iteration, persona-paired against base, $n = 96$, Wilcoxon. The K=5 Q1+Q2 arm ran to iteration 10; every other arm stops at 5.

NOT THIS

Not that K=5 beats K=0 here. The 2.968 arm had five more training iterations than its K=0 sibling. Matched iteration for iteration, K is a wash: mean Δ +0.060 (Q1+Q2), -0.020 (WAI-SR), +0.018 (CSQ-8), with 1 of 15 matched pairs reaching $p < .05$ — which is what chance produces.

The finding that shaped Exp3: two instruments did not train

THE NEGATIVE RESULT
Train PTO on WAI-SR and WAI-SR does not improve. Train it on CSQ-8 and CSQ-8 does not improve. Only Q1+Q2 responds to being optimised.

trained on	measured on	base	best	Δ	p
Q1+Q2	Q1+Q2	2.378	2.968	+0.590	<.0001
WAI-SR	WAI-SR total	2.708	2.788	+0.081	.414
CSQ-8	CSQ-8	2.182	2.255	+0.073	.304

Each arm scored on its OWN training instrument — the fairest possible test of whether the optimisation took. n = 96 paired, uncorrected.

- **Why this matters for the thesis.** It is the reason RQ-iii — “does the oracle questionnaire change the conclusion?” — is still open rather than answered. Exp2 tried, and two of three instruments produced no training signal worth measuring.
- **It is also why Exp3 trains on Q1+Q2 only,** and reads all 8 instruments at evaluation. Choosing what to READ is free; choosing what to TRAIN ON costs a full run each.

Why no Exp2 number appears on an Exp3 axis

THE STANDING WARNING
Exp2 scores 2.378 at base and Exp3 scores 3.008 — same base model, same oracle. The gap is generation precision: Exp2 ran 4-bit NF4, Exp3 bf16.

measured	Exp2	Exp3	vs what the docs claim
Degenerate therapist turns	8.73%	0.34%	reproduces (claimed ≈9.5% vs ≈0.3%)
Base Q1+Q2, all conversations	2.378	3.008	reproduces (claimed 2.38 vs ≈3.0)
Base Q1+Q2, non-degenerate only	2.648	3.022	does NOT (claimed ≈2.93 ≈ parity)

- **Removing degenerate conversations closes 41% of the gap, not all of it.** A clean-vs-clean gap of +0.374 survives every filter tried. The conclusion — never compare levels across experiments — is unchanged and if anything stronger.
- **But the stated MECHANISM is weaker than advertised.** Under the repo's own loop detector the two experiments are 1.4× apart, not 30×. And a second confound was never mentioned: Exp2 stopped generation on one ChatML marker where Exp3 stops on two and post-cleans, leaving 2.2% of Exp2 turns carrying raw control markers against 0% in Exp3.

What Exp2 settled, and what it handed forward

SETTLED

A 1B therapist can be trained by PTO. The Exp1 result was not a property of model size.

It survives harder patients and a schema-constrained oracle.

Which instrument you TRAIN on matters enormously — two of three produced nothing.

Absolute scores are not portable across experiments. Report within-experiment contrasts only.

HANDED FORWARD

Four things changed at once, so nothing here attributes to a cause.

Look-ahead did nothing at matched iterations. Is the Exp1 K result regime-specific, or was Exp2 simply too short and too confounded to see it?

Still one model as both patient and oracle. Still no independent measurement.

PTO has never been compared to anything. Beating an untrained baseline is a low bar — how does it fare against a real alternative optimiser?

ACT III

Exp3 — the controlled comparison

One lever at a time. PTO and GRPO, $K \in \{0, 5\}$, matched MCL, matched candidate budget ($M = G = 8$), matched temperature, one oracle — and a second grader held out.

The design, and the two things Exp1 and Exp2 lacked

- **A comparator.** GRPO — group-relative optimisation. It slices an on-policy rollout and scores G completions per prompt. PTO grows a best-of-M reranked trunk. Same candidate budget, same temperature, same oracle.
- **An independent grader.** Claude Haiku 4.5. Different model family, never played the patient, never was the training reward. Every state scored twice.
- **Everything else held fixed.** MCL = 12, bf16, Q1+Q2 as the only training reward, all 8 instruments read at evaluation.

THE GRID

4 arms $\text{PTO} / \text{GRPO} \times K \in \{0, 5\}$

40 model states $11 + 11 + 11 + 7$, base included

2 graders every state scored on both

30,720 cells $40 \times 8 \text{ rubrics} \times 96 \text{ personas}$, per grader

Complete except GRPO $K=5$, which stopped at iteration 6 — slide 26.

The lever works — in opposite directions

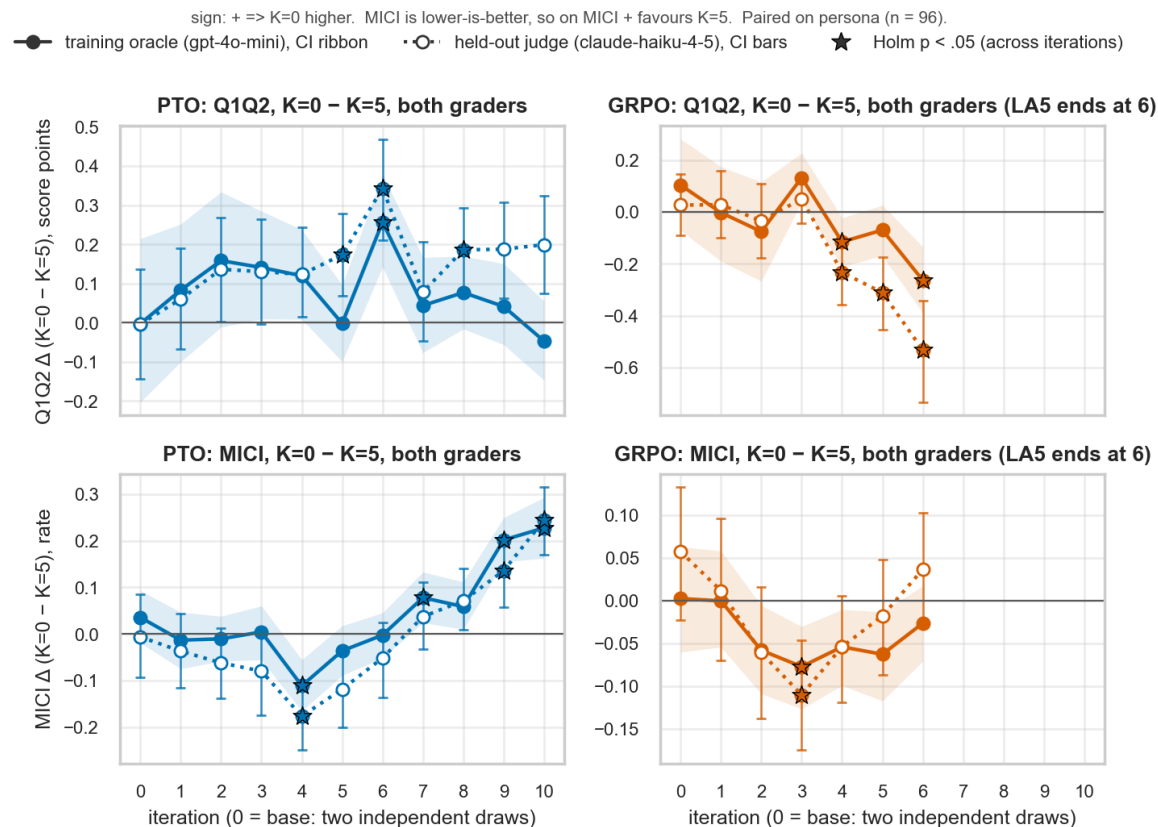
VERDICT
Look-ahead HELPS GRPO and HURTS PTO. At iteration 6 both effects are significant, on both graders.

iteration	PTO · primary	PTO · held-out	GRPO · primary	GRPO · held-out
4	+0.120 (0.20)	+0.123 (0.21)	-0.115 (-0.25) *	-0.233 (-0.37) **
5	-0.002 (-0.00)	+0.173 (0.33) *	-0.070 (-0.13)	-0.311 (-0.43) **
6	+0.257 (0.42) ***	+0.343 (0.51) ***	-0.263 (-0.42) ***	-0.533 (-0.55) ***

Q1+Q2, persona-paired mean difference (Cohen's dz). Sign: + = K=0 higher. * p_holm < .05 ** < .01 *** < .001

AND IT REVERSES EXP1
Exp1 found K=5 above K=0 with PTO. Exp3 finds the opposite for PTO — and finds Exp1's direction only for GRPO, an optimiser Exp1 never ran. The lever's sign is not a property of look-ahead.

The same lever, both optimizers, both graders



READ IT AS

Two panels that mirror each other. Whatever look-ahead is doing, it is not a property of the lever alone — it is an interaction with the optimizer.

The interaction is significant on both graders

VERDICT
The $K \times$ optimizer interaction now reaches significance on the PRIMARY oracle too, not only on the held-out judge.

iter	DiD · primary	dz	p_holm	DiD · held-out	dz	p_holm
4	+0.235	0.286	.104 (ns)	+0.356	0.401	.008
5	+0.068	0.095	1.000 (ns)	+0.484	0.525	<.001
6	+0.520	0.605	<.001	+0.876	0.754	<.001

- **It is not a single-iteration artefact.** On the held-out judge dz runs 0.167, 0.090, 0.401, 0.525, 0.754 over iterations 2 to 6 — it dips at 3, then rises steadily. Iteration 0 — four independent draws of the same untrained policy — sits at -0.033 , this design's noise floor.
- **It holds on every instrument but one.** At iteration 6 the interaction is Holm-significant under both graders on all eight rubrics except MI-inconsistency.

RETIRED
“The grader that WAS the training reward cannot see this effect.” True through iteration 5, false at 6. The defensible claim is now that it sees it $1.7\times$ less sharply ($0.876 / 0.520$).

Does the gain survive a grader that was never the reward?

VERDICT
On GRPO, K=5 keeps 84% of its measured gain under the held-out judge against K=0's 57%, and the intervals are disjoint.

method	iter	K=0 retention	95% CI	K=5 retention	95% CI	disjoint?
GRPO	6	0.567	[0.379, 0.728]	0.844	[0.745, 0.959]	yes
PTO	10	0.823	[0.720, 0.947]	0.639	[0.551, 0.738]	NO – overlapping

- **Retention** = gain seen by the held-out judge ÷ gain seen by the training oracle. 1.0 means the improvement is fully real to a grader with no stake in it; a low number is a gain the training oracle believes and an independent grader does not.
- **Read the PTO row as directional only.** 0.823 vs 0.639 looks like a clean separation and is not — the intervals overlap by 0.018. It IS disjoint on Q2 and MITI, so name the metric.

NOT THIS
Not that PTO reward-hacks and GRPO does not. PTO's K=0 arm retains 0.823 — higher than either GRPO arm. The contrast is WITHIN method, across K.

At K = 0, PTO wins clearly

VERDICT

At the matched 10-iteration endpoint with no look-ahead, PTO beats GRPO on both graders, with a large effect on the held-out judge.

metric	Δ primary	dz	p_holm	Δ held-out	dz	p_holm
Q1+Q2	+0.507	0.729	<.001	+0.609	1.265	<.001
MITI (MI integrity)	+0.352	0.459	<.001	+0.253	0.648	<.001
MICI (inconsistency) ↓	-0.346	-0.989	<.001	-0.225	-0.667	<.001

- **Sign.** Δ = PTO – GRPO. Positive favours PTO; MICI is lower-is-better, so its negative Δ also favours PTO.
- **Mechanism.** GRPO K=0 peaks at iteration 8 (4.082) then regresses to 3.753 while its MI-inconsistency climbs to 0.838. PTO climbs steadily to 4.260.

At K = 5, the answer flips

VERDICT
Turn look-ahead on and GRPO wins. At matched iteration 6 it beats PTO on both graders; best-vs-best it ties on the primary and wins on the held-out judge.

comparison	Δ primary	dz	p_holm	Δ held-out	dz	p_holm
matched iteration 6	-0.332	-0.437	<.001	-0.397	-0.599	<.001
best vs best	+0.078	0.133	1.000 (ns)	-0.168	-0.352	.009

- $\Delta = \text{PTO} - \text{GRPO}$, so every negative number on this slide favours GRPO.
- **So RQ-i and RQ-ii are not separable.** “Which optimizer is better” has no answer that is not conditional on K, and “does look-ahead help” has none that is not conditional on the optimizer. That is the central finding of the thesis.

NOT THIS
Not that GRPO overtakes PTO given enough iterations. GRPO K=5 has SIX; PTO has ten. The flip is at matched iteration, and the arm is censored.

An iteration is not a unit of spend

VERDICT

On the compute axis PTO dominates outright: it reaches iteration 10 for 8.1 GPU-h against GRPO's 27.9 — 3.4× cheaper, and it scores higher.

arm	iterations	GPU-h	GPU-h / iter	Q1+Q2 primary	Q1+Q2 held-out
PTO K=0	10	8.12	0.81	4.260	2.866
PTO K=5	10	19.68	1.97	4.307	2.667
GRPO K=0	10	27.91	2.79	3.753	2.257
GRPO K=5	6 (+ partial 7)	30.53	5.09	4.229	2.903

- **Why PTO is cheap.** Its dominant phase is building preference pairs — 5.7 of 8.1 h — which does not scale with the optimizer step count. GRPO computes its reward inside the training loop, so every step pays.
- **Look-ahead costs ~1.9× per optimizer step,** measured as median ratios 1.97 / 1.96 / 1.91 at iterations 3 / 4 / 5. The two GRPO arms are no longer budget-matched: K=5 is 9.4% more expensive at its last adapter.

RQ-iii

Does the oracle questionnaire change the conclusion?

HELD — AND EXP2 IS WHY

Exp3 trains every arm on Q1+Q2 and EVALUATES on all 8 instruments — so we can already say the conclusions do not depend on which rubric we READ.

Whether they depend on which rubric we TRAIN ON is the open half. Exp2 is the only experiment that tried it, across three instruments — and two of the three produced no measurable gain on their own metric (WAI-SR $p = .414$, CSQ-8 $p = .304$).

So the honest position is not “untested”. It is “tested once, in a confounded regime, with a mostly negative result”. Redoing it properly means four more Exp3 arms at 8–28 GPU-h each.

Recommendation: keep it held. RQ-i and RQ-ii are the thesis.

The measurement thread, in one slide

VERDICT

An independent grader in a different model family reproduces 88.5% of all 6,240 arm × metric contrasts — and 99.5% of the large ones.

subset	n	same sign
all contrasts	6,240	88.5%
$ \Delta \geq 0.10$	4,035	94.5%
$ \Delta \geq 0.25$	2,539	97.4%
$ \Delta \geq 0.50$	1,309	99.5%

— **6,240** = 8 metrics × C(40, 2) = 8 × 780.

- **Oracle repeatability.** ICC(2,1) 0.86–0.99 across four re-draws of the same conversations.
- **Dependability, one judge.** Strong for Q1/Q2/WAI-SR/CSQ-8/PCT (0.91–0.96) and MICI (0.85). MITI is the weak instrument at 0.62 — name it whenever a channel result rests on MITI.
- **This is what Exp1 and Exp2 never had.** In both, the model that produced the reward was also the model that graded the result. Every held-out number in this deck exists only in Exp3.

One honest slide about the run that stopped

WHAT HAPPENED

GRPO K=5's iteration 7 needs 106 optimizer steps. 40 are on disk. Across four Colab sessions, 132 further steps were computed and then thrown away.

session	date	steps trained	steps saved	outcome
1	19 Aug	1 → 103	1 → 30	writes to Drive stopped mid-save
2	20 Aug	—	—	OpenAI org spend limit (384 of 395 log lines)
3	20 Aug	—	—	OpenAI org spend limit
4	20 Aug	31 → 99	31 → 40	writes to Drive stopped mid-save

- **Training was never the problem.** Steps after the stall ran at 162.1 s against 161.1 s before it. GPU and oracle were healthy throughout — only new files stopped reaching Drive.
- **It is not chronic.** All 16 previously completed iterations have exactly as many saved artifacts as optimizer steps. Iteration 7 is the only anomaly.
- **The fix is operational.** Write checkpoints to local Colab disk and copy to Drive once per iteration. Nothing in the experiment name, the reward or the data changes.

Where the evidence is thin — said out loud

- **Every contested endpoint is a single 96-conversation draw.** The only noise floor is at the base: 4 independent draws of the identical base policy give 54 same-policy contrasts, 0 of which reach even uncorrected $p < .05$ (max $|dz|$ 0.128). Reassuring for the base; silent about a trained checkpoint.
- **GRPO K=5 is censored at 6.** Every statement about it is “within six iterations”. Whether its lead persists, or it regresses the way GRPO K=0 did after iteration 8, is unobserved.
- **MITI dependability is 0.62 off one judge,** and MITI carries the MI-integrity channel results. There is no channel-level ICC at all.
- **All 96 personas are used for both training and evaluation** at every iteration, so every number is in-sample with respect to the patient distribution.
- **Exp1 and Exp2 have no held-out grader at all,** so their internal contrasts cannot be checked the way Exp3's can. Act I's re-scoring is one additional grader on the transcripts — not a second draw, and not a second experiment.

What the three experiments establish together

claim	Exp1	Exp2	Exp3	reading
PTO improves a therapist LLM	yes ✓	yes ✓	yes ✓	replicated 3×, 2 model sizes
...and it survives an independent grader	—	—	yes ✓	Exp3 only
Look-ahead helps PTO	weakly	wash	no ✗	sig. on 1 of 4 metrics in Exp1; null in Exp2; reversed in Exp3
Look-ahead helps GRPO	—	—	yes ✓	the direction Exp1 saw, wrong method
The training instrument matters	—	yes ✓	open	2 of 3 Exp2 arms did not train
PTO beats a real alternative	—	—	at K=0	at K=5 it loses

— **The one claim that strengthened across all three experiments is the plainest one.** PTO works. It replicates across two therapist model families, two patient regimes, two oracles, and — only in Exp3 — a grader with no stake in the outcome.

— **The one that dissolved is the headline of the published paper.** Look-ahead is not a knob with a sign. It helps group-relative optimisation and hurts preference-tree optimisation, and Exp1 could not have seen that because it had only one optimiser.

Three things I would like decided today

1

Finish GRPO K=5?

Resume iteration 7 and run to 10: ~16–18 GPU-h, ~\$85–130. The arm lands near 48–50 GPU-h against its K=0 sibling's 28.

Recommendation: yes — it is the only arm without an endpoint, and it is currently the best-scoring arm on the held-out judge.

2

How much of Act I goes in the thesis?

The re-audit found the published ANOVA was run on 3 groups of 15, and the depth contrast is significant on one metric of four. The re-scoring supports the direction but not the strength.

Recommendation: include it. A thesis that re-examines its own prior work is stronger than one that cites it — and Exp3 reverses the sign for PTO regardless.

3

Buy the replicate draw?

A second independent 96-conversation draw of ~5 contested endpoints: ~\$10 and about one A100-hour, or four free hours locally. No code change needed.

Recommendation: yes — it either retires the endpoint-fragility objection thesis-wide or tells us the headline is fragile.

IN ONE LINE

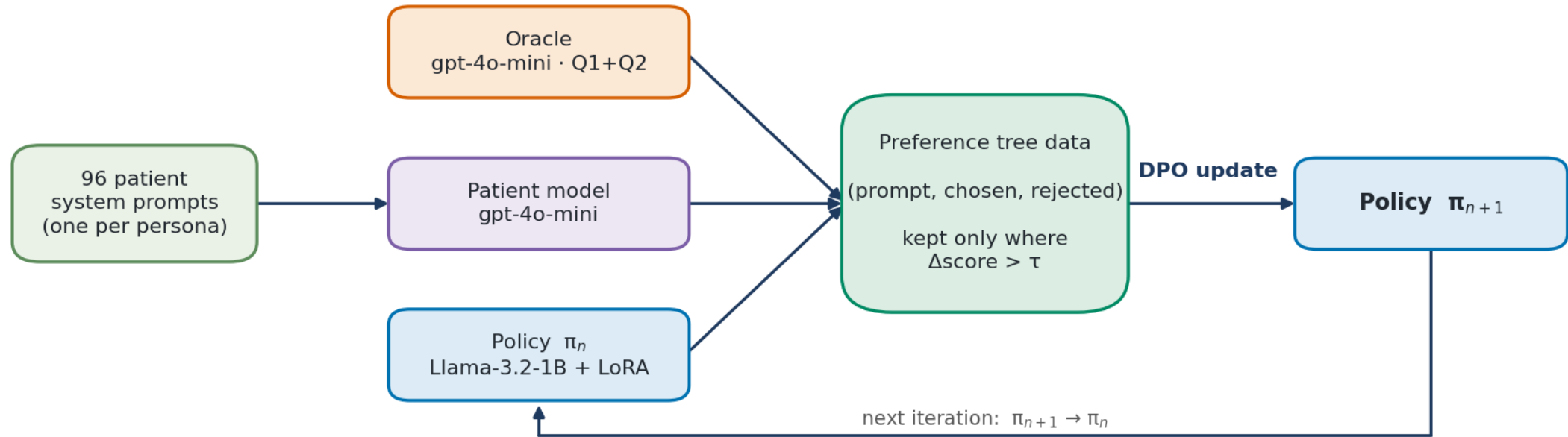
Look-ahead is not a knob with a sign.

It helps group-relative optimization and hurts preference-tree optimization, and which optimizer you should prefer depends on whether you turned it on.

Three experiments agree that PTO works. Only the third could have found that the lever Exp1 was built around points the other way for the method Exp1 used.

PTO remains 3.4× cheaper and wins outright at $K = 0$.

PTO — preference tree + DPO

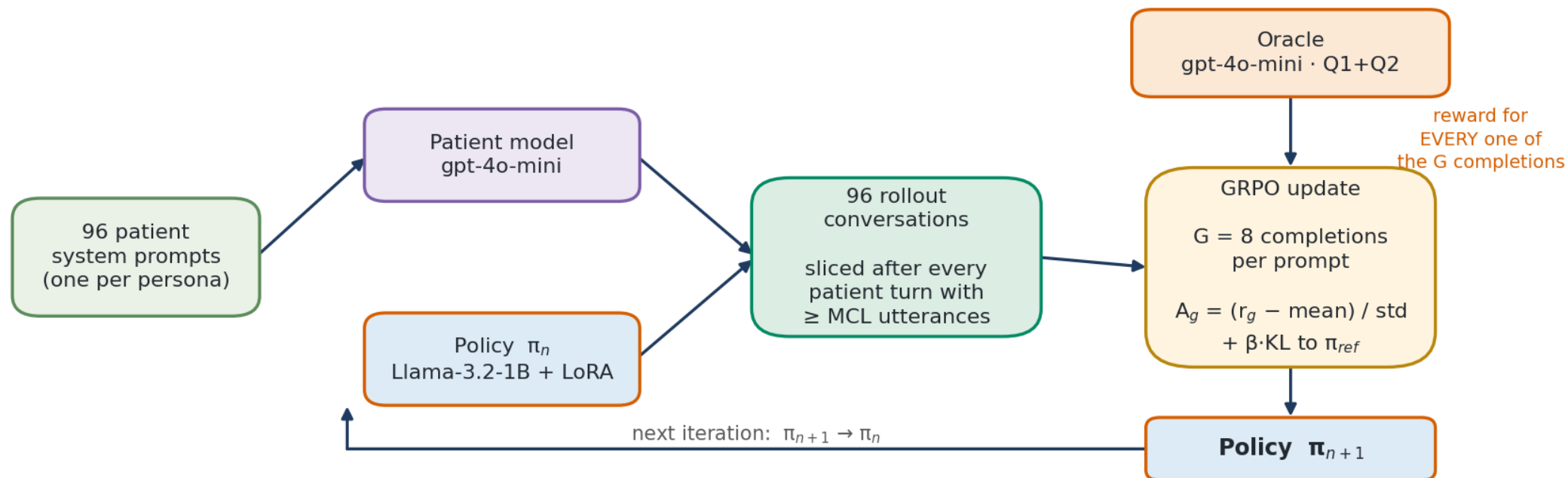


Green = data this iteration produces · purple = an API model we call · blue = the policy being trained · orange = the oracle
The 96 conversations generated at the start of the iteration are also the evaluation set for π_n — there is no separate evaluation pass.

IN ONE LINE

Branch M candidates per therapist turn, look ahead K, score, keep (chosen, rejected) where the margin clears τ , then a DPO update. This is the method Exp1, Exp2 and Exp3 all run.

GRPO — group-relative rollout



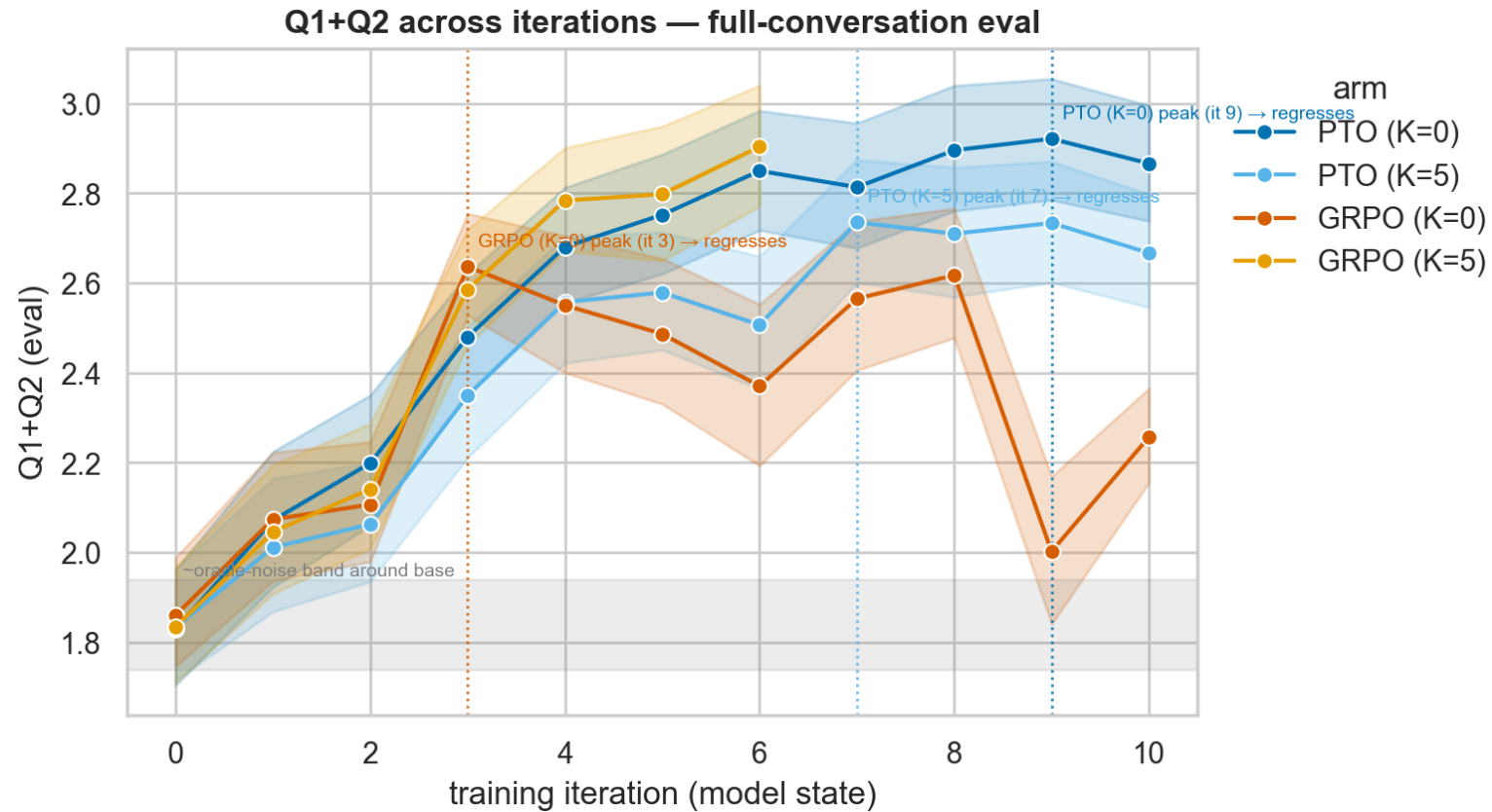
Same colour code as the PTO framework figure. The one structural difference is where the oracle sits: inside the update, not before it.

As in PTO the rollout conversations double as the evaluation set for π_n — but the prompt list is fixed for the iteration, every prompt trains, and nothing is discarded.

IN ONE LINE

Slice the rollout after every patient turn, sample G completions per prompt, score all of them, use the group-relative advantage. The oracle sits INSIDE the update. Exp3 only.

All four Exp3 arms, held-out judge



READ IT AS

GRPO K=0 (orange) is non-monotonic — 2.637 at iteration 3, back to 2.617 at 8, then down to 2.257 by 10. GRPO K=5 climbs to 2.903 in six iterations, the highest last-scored value of any arm on this grader — but it stopped there rather than finishing.

GPU-hours per iteration, reconstructed from artifact mtimes

